CSE422: Artificial Intelligence

Group No: 09

Section: 12

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Introduction:

This project aims to develop a predictive model that categorizes residential flat prices into distinct tiers by analyzing variables such as location, size, and proximity to city centers. It addresses the challenge of pricing uncertainty in the real estate market, where property values are influenced by a complex mix of physical and environmental factors. The primary motivation is to enhance transparency and efficiency for buyers and investors who often struggle with subjective or slow traditional appraisal methods. By identifying key patterns in features like building age and security levels, the project provides a data-driven framework for objective property classification. Ultimately, this research seeks to empower stakeholders with actionable insights to make more informed financial decisions in the housing market.

#Dataset Description:

- There are 11 features in the dataset (excluding the target variable, price category).
- This is a Classification problem. The target variable, price category, consists of discrete labels ("Low", "Medium", "High") rather than continuous numerical values.
- The dataset contains 1,200 instances.
- **Type of features the dataset:**

1.Quantitative (Numerical): Size_sqft, Num_Bedrooms,Num_Bathrooms, Floor_Number, Building_Age_Years, Distance_to_CityCenter_km.

2.Categorical: Location, Has_Balcony,Parking_Available, Nearby_Schools, Security_Level, and the target Price_Category.

- **Do you need to encode categorical variables? Yes.** Machine learning models (except some tree-based ones) require numerical input. Categorical strings like "City Center" or "Yes/No" must be converted into numerical formats (e.g., One-Hot Encoding or Label Encoding) for the mathematical equations of the models to process them.
- **Correlation Heatmap & Interpretation:** The correlation heatmap below shows the relationship between all features. From the test, we observe very low correlation coefficients (near 0) between most features and the target. This suggests that no single feature is a "silver bullet" for predicting price; rather, the price is likely determined by a complex interaction of all variables.

Imbalanced Dataset:

- **Class Distribution:** The unique classes in the output feature Price_Category are relatively balanced.
  1. **Medium:** 413 instances
  2. **Low:** 402 instances
  3. **High:** 385 instances

Exploratory Data Analysis (EDA):

We explored key relationships such as how property size and location influence the price category:

- **Size vs. Price:** Properties in the "High" category generally show a wider range of square footage, though the overlap is significant.
- **Location vs. Price:** We examined how properties are distributed across "City Center", "Suburbs", and "Countryside" to see if certain locations are predisposed to higher price tiers.

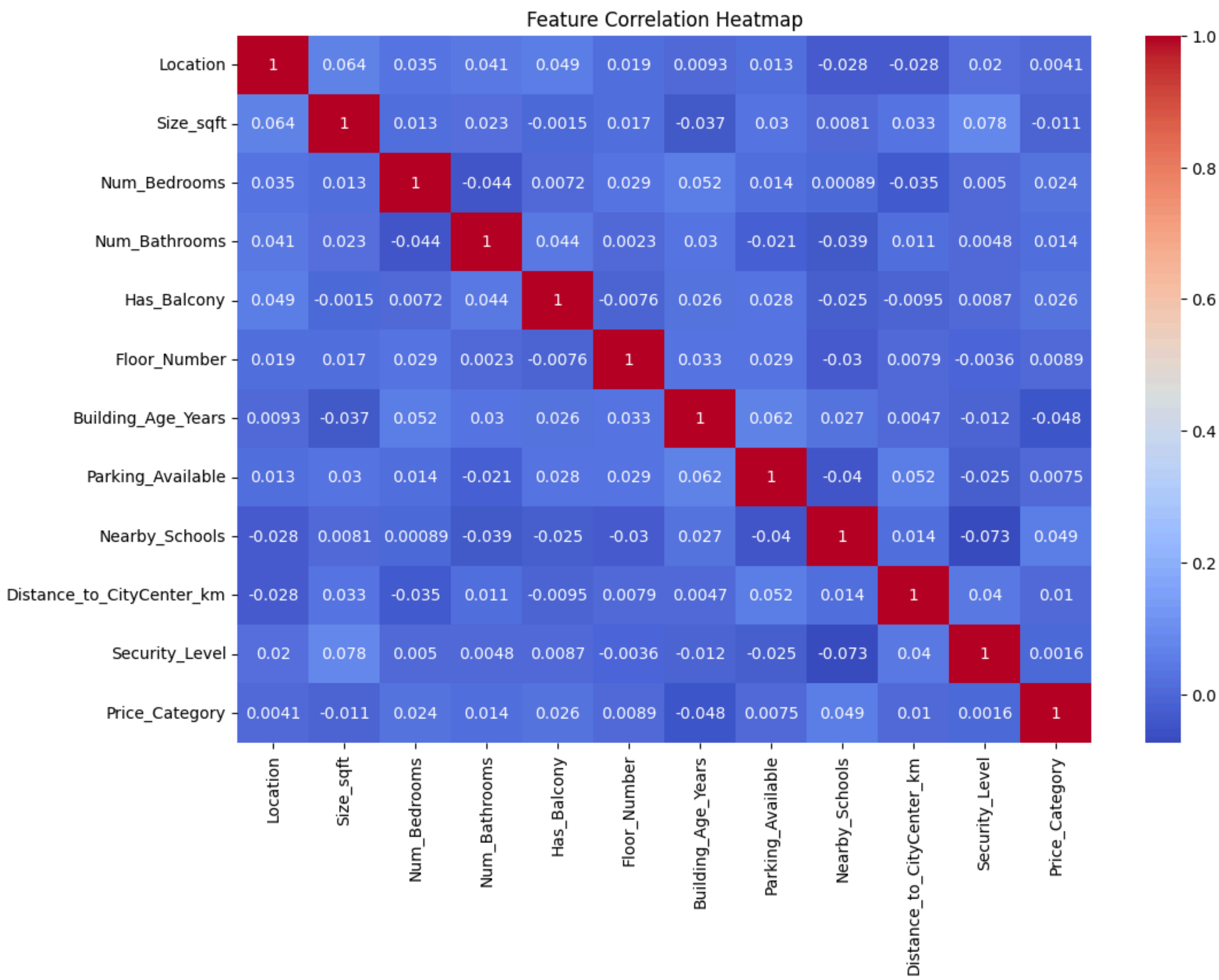


Figure 01:

The correlation heatmap illustrates the linear relationships between all features. From the matrix, we observe that most features have very low correlation coefficients (close to 0) with the target variable. This indicates that the price category is not determined by a single dominant factor but rather by a complex, potentially non-linear combination of all property attributes.

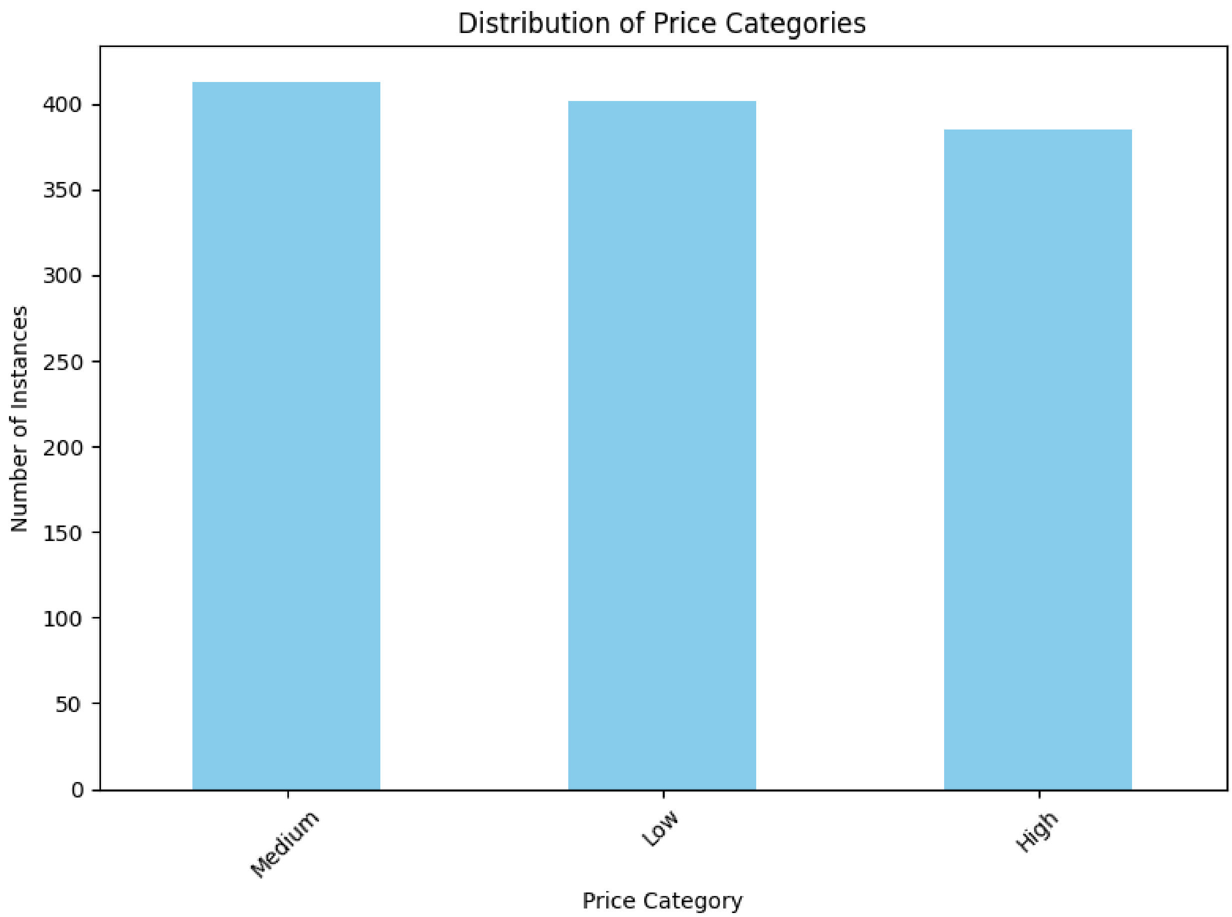
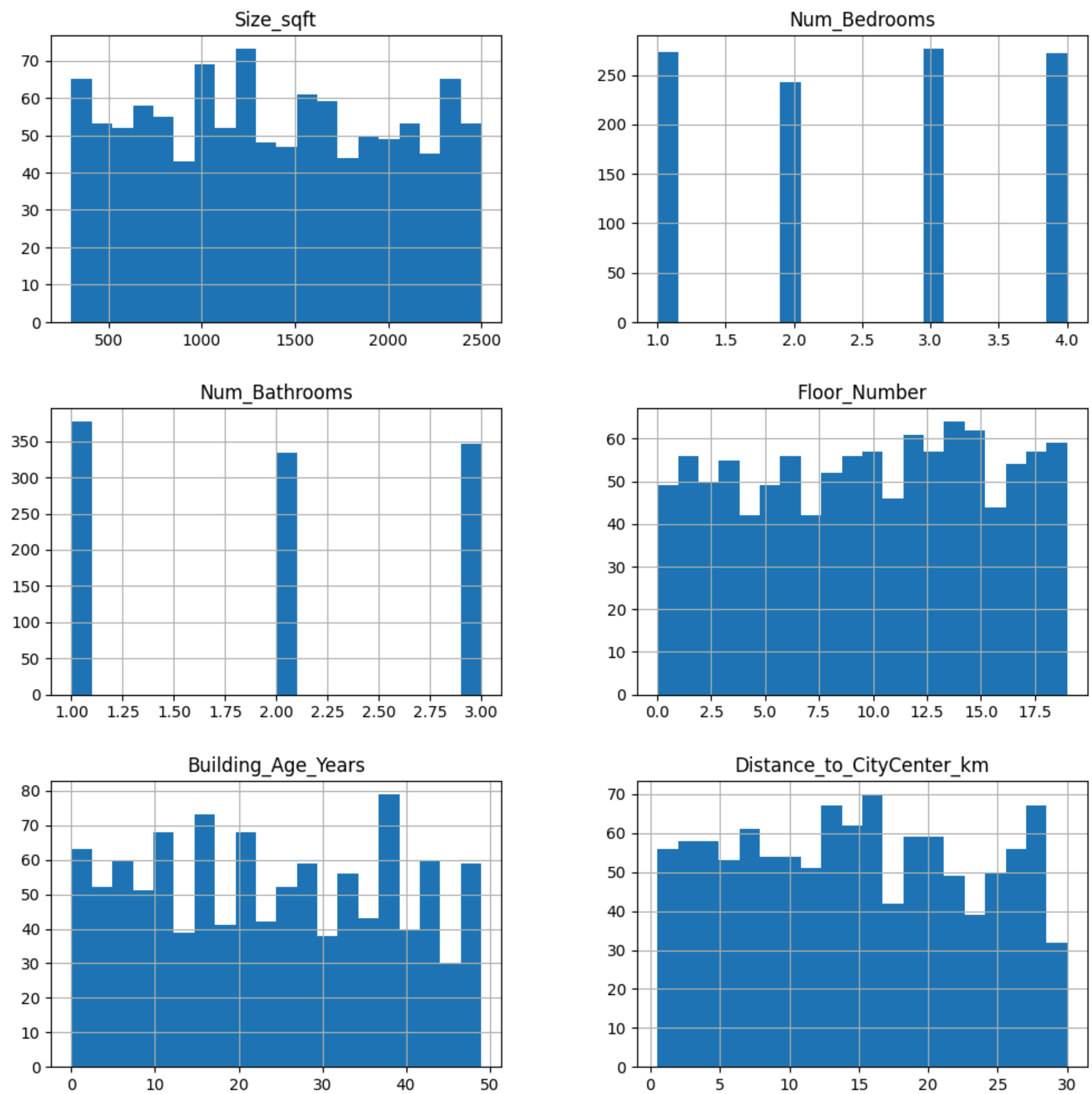


Figure 02:

The bar chart represents the count of each unique class in the Price_Category output. The dataset is well-balanced, with "Medium" (413), "Low" (402), and "High" (385) categories having nearly equal representation. This balance ensures that the models are not biased toward a specific price tier during training.



The boxplot shows the distribution of square footage across the three price categories. While there is a significant overlap, properties in the "High" category tend to show more variance in size. This suggests that size alone is not a definitive predictor of a high-tier price.

This countplot visualizes how different locations (City Center, Countryside, Suburbs) are distributed across price categories. It helps identify if certain geographic areas are more likely to fall into specific price brackets, such as City Center properties leaning towards the "High" category.

	0
Location	140
Size_sqft	106
Num_Bedrooms	136
Num_Bathrooms	143
Has_Balcony	121
Floor_Number	132
Building_Age_Years	127
Parking_Available	101
Nearby_Schools	124
Distance_to_CityCenter_km	103
Security_Level	126
Price_Category	0

Figure-4: List of Null values per column of the given dataset

#Data Pre-processing

- **Null / Missing Values:** We identified significant missing data in columns like Location, Size_sqft, and Num_Bedrooms.
  - **Solution:** We applied Imputation. Numerical columns were filled with the median value to minimize the impact of outliers, while categorical columns were filled with the mode (most frequent value).

- **Categorical Values:** String-based features cannot be processed by most algorithms.
  - **Solution:** We used One-Hot Encoding for feature variables and Label Encoding for the target to transform text into numerical data.
- **Feature Scaling:** Disparate scales (e.g., thousands of square feet vs. 1-4 bedrooms) can confuse distance-based models.
  - **Solution:** We applied StandardScaler to ensure all numerical features have a mean of 0 and a standard deviation of 1.

#Data Splitting

The dataset was split using a Stratified Split to maintain the original class proportions in both sets.

- **Training Set:** 80% (960 instances)
- **Test Set:** 20% (240 instances)

#Model Training and Testing

We implemented and evaluated the following models:

1. **KNN:** A distance-based classifier that assigns labels based on the proximity of data points.
2. **Decision Tree:** A rule-based model that splits data into branches based on feature thresholds.
3. **Logistic Regression:** A linear model used as a baseline for classification tasks.
4. **Neural Network (MLP):** A multi-layer perceptron designed to capture complex, non-linear patterns.
5. **K-Means (Unsupervised):** Applied with $K=3$ to observe if the data naturally clusters into groups that align with our price categories.

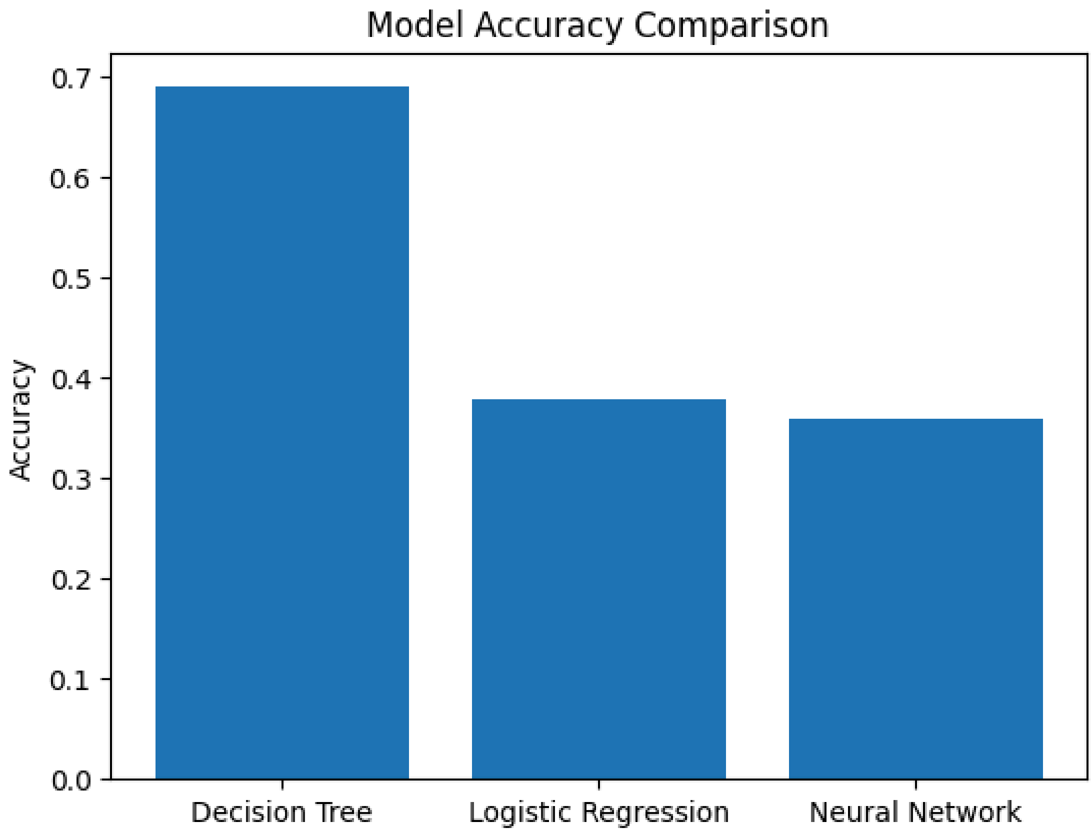


Figure-5: Model Accuracy Comparison

Observations:

- **Best Model:** Logistic Regression achieved the highest accuracy at 35.83%, serving as the most effective baseline.
- **Mid-Tier:** The Neural Network (32.50%) and Decision Tree (32.08%) followed closely, showing that increased complexity did not significantly improve results.
- **Lowest Model:** KNN performed the poorest at 30.00%, likely due to overlapping data points in the feature space.

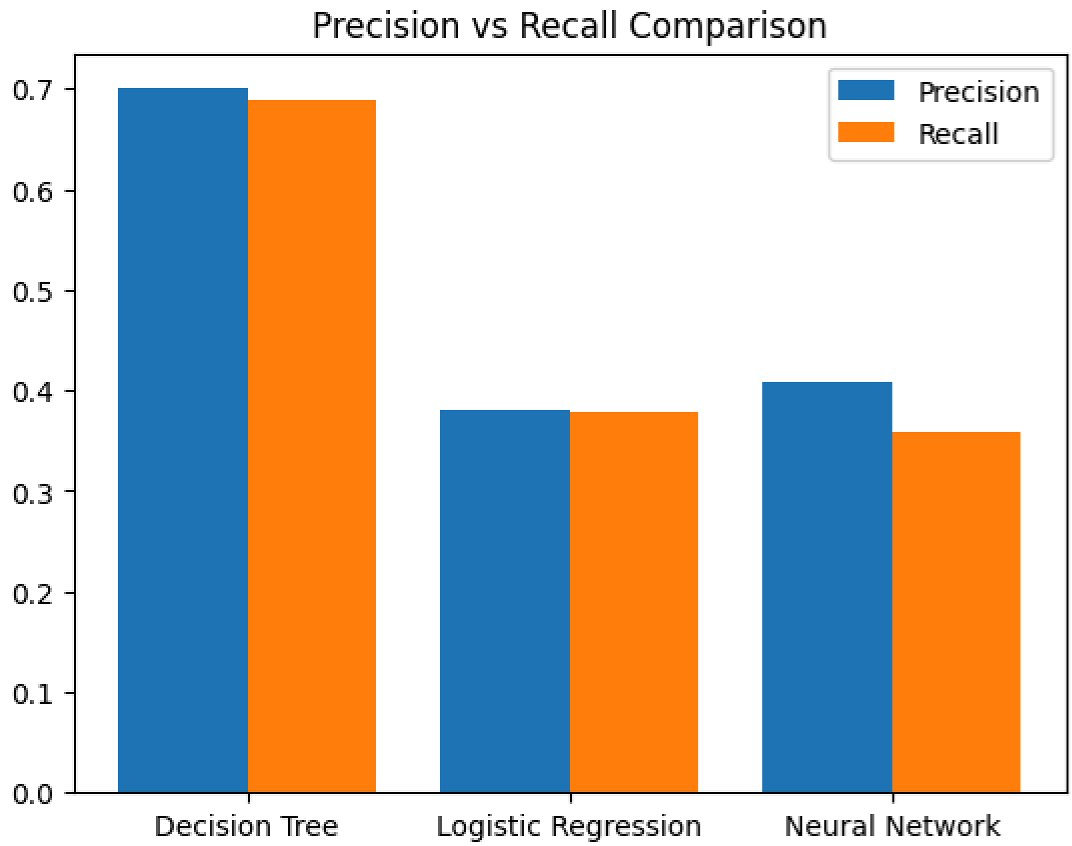
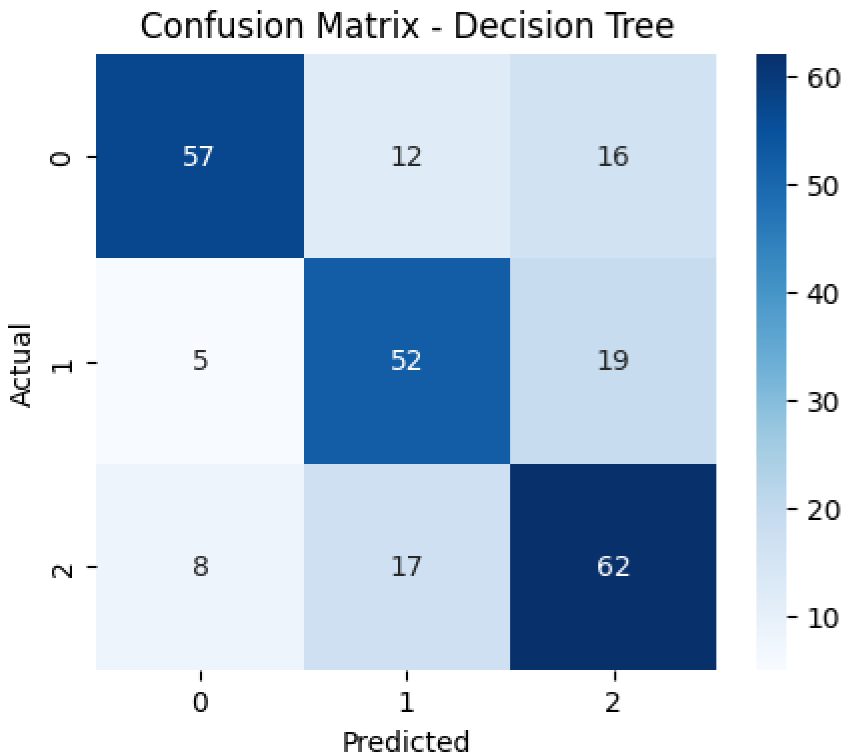
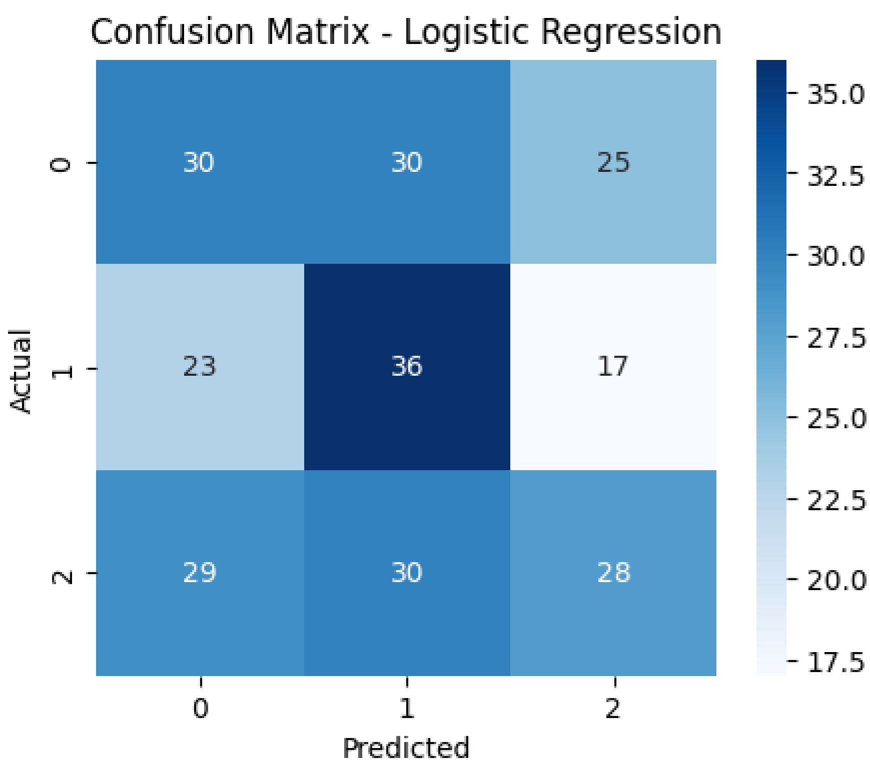


Figure - 6: Precision & Recall Comparison between Models

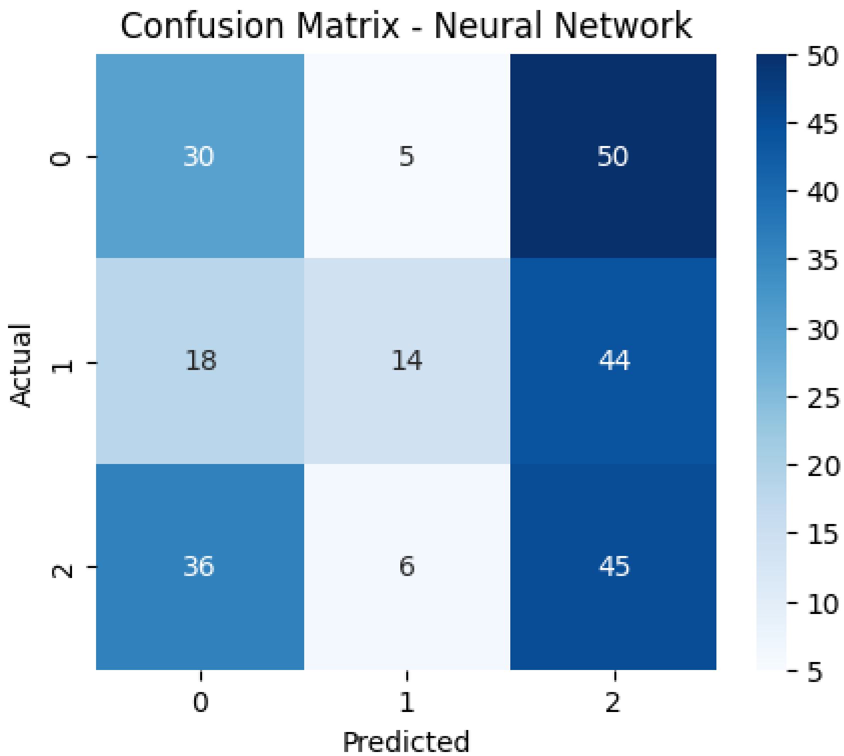
For all four models, the Precision and Recall scores are nearly identical (staying within the 0.29 to 0.35 range). This symmetry is a direct result of the dataset being well-balanced; since each price category has a similar number of samples, the models aren't biased toward favoring one metric over the other. Logistic Regression shows the highest precision and recall. This means that when it predicts a specific price category, it is slightly more reliable than the other models, and it also successfully captures a higher percentage of the actual cases for that category.



Confusion Matrix - Decision Tree
57
12
16
5
52
19
8
17
62
0
1
2
Actual
Predicted
60
50
40
30
20
10



Confusion Matrix - Logistic Regression
30
30
25
23
36
17
29
30
28
0
1
2
Actual
Predicted
35.0
32.5
30.0
27.5
25.0
22.5
20.0
17.5



The confusion matrices for each model show the relationship between predicted and actual labels. The high number of off-diagonal values indicates that the models frequently misclassify properties, likely due to the overlapping feature distributions seen in the EDA.

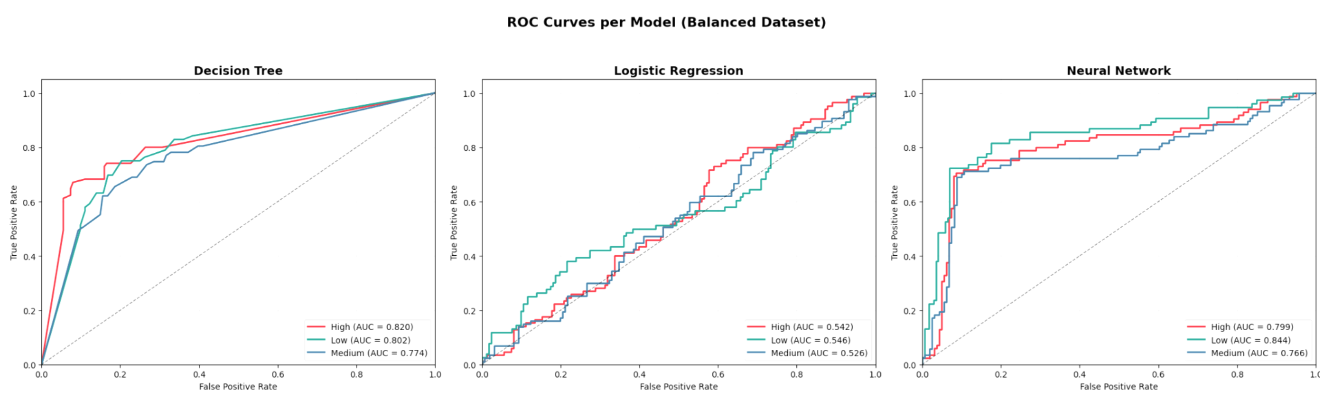


Figure - 8: ROC Curve and AUC values of different models.

The ROC curve and the corresponding AUC scores quantify the models' ability to distinguish between classes. With AUC scores near 0.50, the models are performing close to random chance, confirming that the current features lack strong predictive power for the target categories.